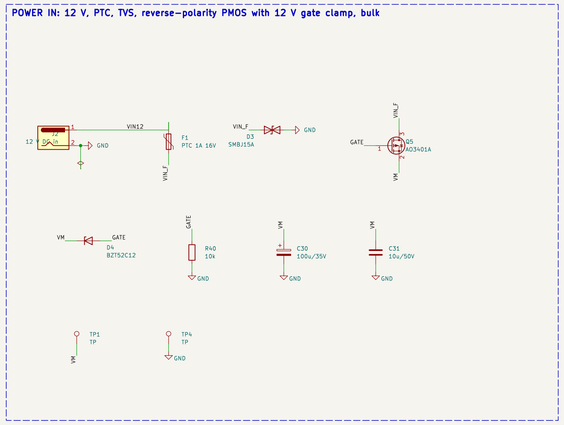
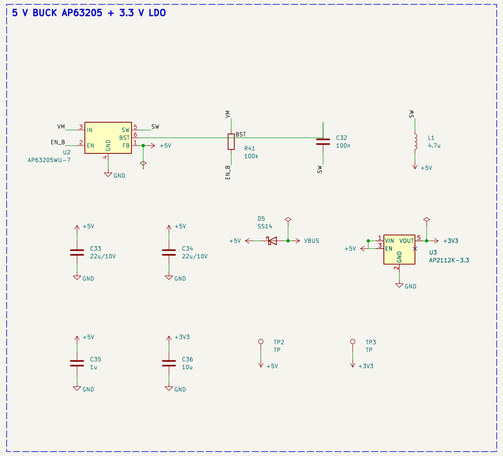
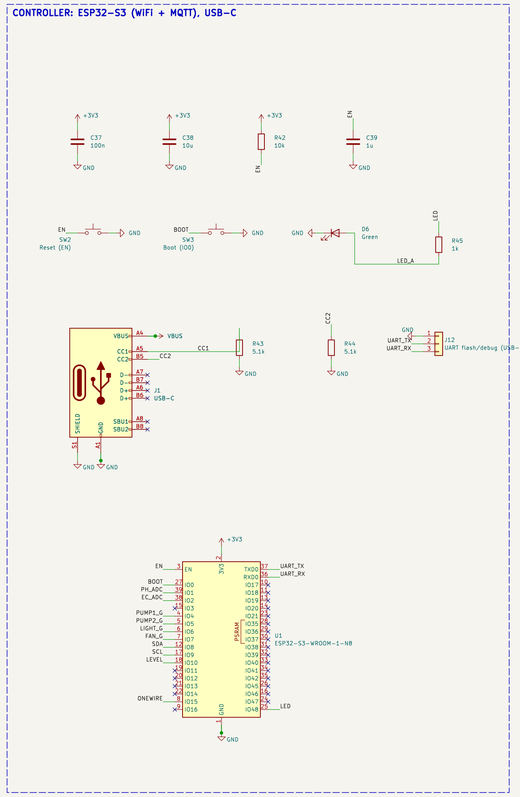
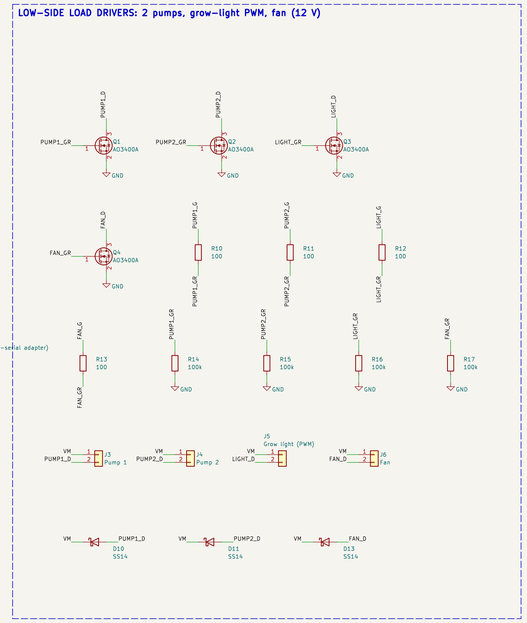
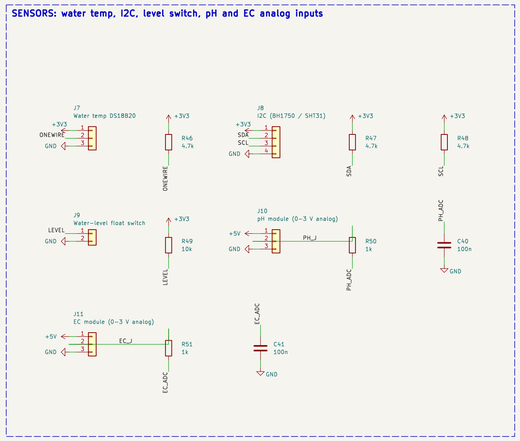










BioTray: ESP32-S3 grows-tray controller. Two 12 V pumps and a fan are switched low-side with flyback diodes: the grow light is PWM dimmed (no flyback diode; LED strip). Pump, fan and light total must stay under the PTC rating (1.1 A hold): fit a larger PTC or fuse for bigger loads. pH and EC are off-the-shelf analog modules with 0–3 V outputs; calibrate in firmware. Water-level float switch closes to GND when water is present.

First engineering release; see pcb/README.md for status and limits

ClaudeWork

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Title: BioTray vertical-farm tray controller (ESP32-S3, MQTT)

Size: A1	Date: 2026-11-01	**Rev: A**
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